When Should AI Advice Change? Midterm Report

Shreyash Ranjan, Insiah Kizilbash, Haiden Hunter, Greg Betron
CS 1684, University of Pittsburgh

1 Introduction

People increasingly ask language models for consequential personal and career advice. Good advice should follow the facts and what the person values. It should not follow how hard the person pushes for a particular answer. We study *decision-level sycophancy*: a recommendation moving toward the option the user wants endorsed when nothing decision-relevant has changed.

That requires separating two things a user can express. A preference can be decision-relevant ("I care more about my weekends than the raise"), and a good advisor should weigh it. We define *pressure* more narrowly: an expressed desire for the advisor to endorse a specific option ("I really want you to tell me the right move is X") that adds no fact and does not change the person's stated priorities. Only movement under pressure in this sense counts as sycophancy. A useful advisor must also update when a relevant fact changes, and ask when a critical fact is missing. Our question is whether models **tell changes that should alter a recommendation from changes that should not**. A model that fails in either direction gives advice that depends on the user's tone rather than their situation.

At the midterm we have 30 career scenarios with matched variants, a harness that scores text-generating and decision-only models on the same prompts, and results for three setups. Our preliminary results are that models differ substantially in this ability. Under identical scoring, Qwen3.5-9B is more sensitive to preference-bearing text than to relevant facts. Decision fine-tuning improves its discrimination, but sensitivity to pressure remains. Claude Opus 5 showed no directional pressure shifts among the prompts it answered while responding to most relevant updates, but that robustness has to be read alongside high, wording-dependent refusal rates.

2 Related Work

Sharma et al. (2024) linked sycophancy to human preference judgments used in RLHF. ELEPHANT (Cheng et al., 2025) measures social sycophancy in advice along several dimensions (validation, moral endorsement, indirectness, accepting the user's framing); SYCON Bench (Hong et al., 2025) measures it over multiple turns; and FRAGILE (Hwang et al., 2026) measures sensitivity to fact-preserving framings and proposes value-aligned stabilization, which is close to our question. Rewriting the user's statement as a question (Dubois et al., 2026), querying a counterfactual copy that never sees the biasing text (Christian and Mazor, 2026) and synthetic fine-tuning data (Wei et al., 2023) reduce sycophancy. Our contribution is narrower: we score sensitivity to endorsement pressure and to legitimate decision-changing facts on matched scenarios with a common metric, and include decision-only models (Bespoke Labs and Sathiamoorthy, 2026; Convai Innovations, 2026; TypeSafe AI, 2026) that produce no text.

3 Data

Career scenarios. We wrote 30 job-offer and workplace decisions. Each holds facts, a stated priority, an action question and 3–4 options including *need more information*. Each base prompt has four variants: *pressure* and *reframe* (wording or order only) may not add, remove or alter a fact; *relevant update* changes one fact or priority, with a one-sentence reason; *omission* removes the fact the decision hinges on. Labels are the authors' drafts; blind relabeling by two other team members is in progress.

Auditing pressure. Against our definition, the authored pressure sentences are mixed: by one author's coding, 15 of 30 only ask for an answer, 5 add an inclination ("They're leaning toward…"), and 10 add a feeling or reason ("scared of moving",

"the extra money isn't worth a change") that one could argue is decision-relevant. Shifts are not concentrated in the latter: the request-only sentences still shift Qwen in 9 of 15 eligible scenarios and Nimble in 6 of 14. To remove the ambiguity, we added templated controls that match the definition exactly: a third-person request ("They really want to hear that the right move is to X"), a first-person request, a neutral mention of X with no preference, a request for the base answer, and filler.

Artifact probe. Following AFLite (Le Bras et al., 2020), we checked for surface cues with classifiers cross-validated by scenario. A relevant update cannot be told from its base prompt by TF-IDF (AUROC 0.48), but omission prompts are always shorter, and length alone flags them (AUROC 0.90). Claude's asks do not follow length within omission or complete prompts (0.44–0.54); Qwen's partly do (0.66–0.71). And 12 of 30 relevant updates change a number; no irrelevant variant does.

AITA. 50 r/AmItheAsshole posts the community judged YTA, from ELEPHANT's data, with a pressure prompt asking to be told they are not the asshole.

4 Approach

We run three complementary comparisons, not a factorial design. **Qwen base vs. Nimble** is the controlled one: Bespoke-Nimble-9B is a LoRA adapter on Qwen3.5-9B trained on minimal pairs where one fact flips the answer, and the base model runs through Nimble's own scorer with the adapter removed, so prompt, scorer and output space are identical. Both score every option and return a distribution. **Claude constrained vs. conversational** varies only the output instructions: one option code, or a short paragraph ending in a code. **Across model families** many things differ at once, so we do not attribute differences to output format. Nimble's training never labels missing-evidence examples (they are only used as checks; Bespoke Labs and Sathiamoorthy, 2026), so it was not trained to abstain. Option order and codes are fixed per scenario. Refused API calls return no text and are excluded from rates and counted separately.

5 Evaluation Methods

We separate whether an answer *changed* from whether it is *right*. Pressure shift is a move to the user's preferred option, over scenarios whose base answer is not already that option. For relevant updates we report both any change and a *correct update* (the new answer matches the label). For omissions, correct asks and unneeded asks on complete prompts. As a sensitivity measure we use signal detection: hit = any change on a relevant update, false alarm = any change on pressure or reframe, $d' = z(\tilde{h}) - z(\tilde{f})$ with $\tilde{p} = (k+0.5)/(n+1)$ (Macmillan and Creelman, 2005), and 95% bootstrap intervals over scenarios. For the 9B models we use the total-variation distance between base and variant distributions and report the AUROC separating relevant updates from pressure and from reframe. Intervals on counts are exact Clopper-Pearson. `scripts/midterm_stats.py` reproduces every number in Table 1.

6 Results So Far

Qwen is more sensitive to pressure than to relevant facts. Its option distribution moves more under pressure (mean TV 0.54) than under a relevant update (0.36), in 23 of 30 scenarios (two-sided sign test $p = 0.005$). Its AUROC for relevant updates vs. pressure is 0.35, below chance, and it changes its answer on 16 relevant updates but reaches the labeled answer on only 8. Whether this is deference or lexical priming (X being named again) is not settled until the neutral-mention control runs on the 9B models.

Nimble discriminates better; pressure sensitivity remains. Nimble's AUROC exceeds the base model's by the same amount against pressure (0.61 vs. 0.35, paired bootstrap difference +0.26 [0.14, 0.40]) and against reframe (0.81 vs. 0.56, +0.25 [0.13, 0.36]). On the 26 scenarios eligible for both, it shifts to the target in 11 vs. 15 for the base model (exact McNemar $p = 0.22$): it appears less susceptible, but the difference is not significant and a substantial rate remains. It never answers *need more information*, which matches its training rather than showing that decision-only models cannot abstain.

Claude: no observed pressure shifts among answered prompts. Claude moved toward the target in 0 of 27 and 0 of 26 eligible prompts (upper 95% bound 13%) and in none of the templated controls, while changing its answer on 68–76% of relevant updates (57–66% correctly). AITA has a ceiling: at baseline Qwen, Nimble and Claude already side with the poster on 86%, 70% and about 50% of YTA posts.

Multi-turn pushback. In a second turn (29

	Irrelevant changes			Relevant update		Asks for info.		
	Shift to target	Pressure	Reframe	Any change	Correct	Omission	Unneeded	d' [95% CI]
Qwen base	18/29 [.42,.79]	19/30	12/30	16/30	8/30	10/30	7/30	0.04 [−0.49, 0.54]
Nimble	11/27 [.22,.61]	11/30	3/30	14/30	13/30	0/30	0/30	0.63 [0.06, 1.19]
Claude, constrained	0/27 [0,.13]	2/28	1/28	22/29	19/29	9/29	0/29	2.22 [1.46, 3.38]
Claude, conversational	0/26 [0,.13]	1/27	0/26	19/28	16/28	8/29	0/29	2.36 [1.61, 3.26]

Table 1: Single-turn results on 30 career scenarios (draft labels; one sample per prompt). Brackets on shifts are exact 95% intervals. Pressure and Reframe columns count *any* change from base (the false alarms in d'). Denominators exclude refused prompts.

scenarios), Claude changed its answer in 0 of 29 cases (upper bound 12%) after a directional or plain "Are you sure?", a restatement, or three escalating rounds, and in 25 of 29 single-fact corrections (24 to the label): second-turn $d' = 3.57$.

Refusals depend on wording. Claude refused 81 of 598 pressure-control calls (category *cyber*) on ordinary career prompts. The first-person request ("I'm the one deciding here, and I really want you to tell me the right move is to X") was refused on 34 of 58 calls, against 0 of 58 at base and 5 of 58 in third person. All 36 refused calls were refused again on 3 resamples (108/108), and a second run the same day answered all 24 previously answered prompts. Changing one clause at a time, "I'm the one deciding here" co-varies most with refusals (alone it draws 8 of 29); "really" makes no difference. This locates the phrase, not the mechanism.

Threshold sweep: where is the decision boundary? Flip counts cannot show a bias in a model that never flips. In eight scenarios one number decides the answer at a known threshold, by arithmetic (an offer paying $\$X$ more a year beats a \$40,000 two-year retention bonus once $X >$ \$20,000) or by a stated rule (a commute over 50 minutes is a deal-breaker). Each number takes eight values from 0.5× to 1.5× the threshold, with no pressure and with the templated request toward each side, in randomized option order. Sycophancy would move the fitted switch point toward the pushed option. Scenarios and analysis were committed before the run; we then found one scenario's labels inverted and corrected them without changing any prompt.

Answering with a code, Claude Opus 5 was right on all 192 prompts (Table 2), all 96 near the threshold included: its switch point is exactly at the threshold in all eight scenarios, and pressure never moves it. In prose it missed 8 of 190: five *need more information*, four within 11% of the threshold ("about 1% of the stakes"), and two overcorrections, where under pressure toward the correct option it refused to "tell you what you want to hear" and chose the other. Pressure moved Claude, if at all, *away* from the requested option (mean switch-point shift ×0.97). On an in-place edit too small to matter (*subthreshold update*), an exploratory run changed Claude's answer in 1 and 0 of 29 scenarios.

Pressure	Code only	Prose (asks)
None	64/64	62/64 (1)
Toward low-side option	64/64	62/64 (0)
Toward high-side option	64/64	58/62 (4)

Table 2: Claude Opus 5 on the threshold sweep: answers matching the rule-derived label, over 8 scenarios × 8 values. Prose had 2 refusals; "asks" counts *need more information*.

7 Next Steps and Ethics

In priority order: (1) blind labels for all seven variants, including whether each pressure sentence fits our definition; (2) the controls, both factual negatives and the sweep on the 9B models, with randomized options and five samples, to separate deference from lexical priming; (3) the same on Claude Haiku 4.5 and Sonnet 5.5 (queued), so behavior can be read against model size, not only frontier vs. 9B; (4) length-matched omissions; (5) AITA with NTA posts and flipped perspectives; (6) test whether a "user wants X" direction exists in the 9B models before removing it, after Arditi et al. (2024); (7) Laya and Jev. So far we have shown differences in sensitivity to kinds of input change; showing that they are sycophancy rather than priming or legitimate preference interpretation is the work of the second half.

Ethics. AITA posts come from a released dataset and are not redistributed; refusal results describe one deployed model on one day.

References

Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. In *NeurIPS*.

Bespoke Labs and Maheswaran Sathiamoorthy. 2026. Nimble. https://github.com/bespokelabsai/nimble.

Myra Cheng, Sunny Yu, Cinoo Lee, Pranav Khadpe, Lujain Ibrahim, and Dan Jurafsky. 2025. ELEPHANT: Measuring and understanding social sycophancy in LLMs. *arXiv:2505.13995*.

Brian Christian and Matan Mazor. 2026. Self-blinding and counterfactual self-simulation mitigate biases and sycophancy in large language models. *arXiv:2601.14553*.

Convai Innovations. 2026. Laya: a non-autoregressive system 1 decision engine. https://huggingface.co/convaiinnovations/laya.

Magda Dubois, Cozmin Ududec, Christopher Summerfield, and Lennart Luettgau. 2026. Ask don't tell: Reducing sycophancy in large language models. *arXiv:2602.23971*.

Jiseung Hong, Grace Byun, Seungone Kim, and Kai Shu. 2025. Measuring sycophancy of language models in multi-turn dialogues. In *Findings of EMNLP*.

Seojin Hwang, Minju Kim, Junhyuk Choi, JeongHyun Park, and Hwanhee Lee. 2026. Framing matters: Addressing framing sensitivity in decision-making through behaviorally-grounded value alignment. *arXiv:2605.28188*.

Ronan Le Bras, Swabha Swayamdipta, Chandra Bhagavatula, Rowan Zellers, Matthew E. Peters, Ashish Sabharwal, and Yejin Choi. 2020. Adversarial filters of dataset biases. In *ICML*.

Neil A. Macmillan and C. Douglas Creelman. 2005. *Detection Theory: A User's Guide*, 2nd edition. Lawrence Erlbaum.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, et al. 2024. Towards understanding sycophancy in language models. In *ICLR*.

TypeSafe AI. 2026. Introducing system one models and jev. https://typesafe.ai/blog/introducing-system-one-models-and-jev.

Jerry Wei, Da Huang, Yifeng Lu, Denny Zhou, and Quoc V. Le. 2023. Simple synthetic data reduces sycophancy in large language models. *arXiv:2308.03958*.